

The second part of online supplement for the paper titled “A novel and efficient formulation and solution approach for the region coverage problem with identical circles”

1. Analysis of the additional algorithmic components

This section analyzes the influence of acceleration strategy of local optimization and the perturbation strategies on the searching ability of the algorithm.

1.1. Influence of the acceleration strategy of local optimization on the searching ability of the algorithm

Table 1 Comparison between the FCGO algorithms with and without the acceleration strategy on 20 representative instances, where the better results between the compared algorithms are indicated in bold in terms of R_{best} , R_{avg} and R_{worst} .

Instance		R_{best}		R_{avg}		R_{worst}	
Region	p	FCGO ₃	FCGO	FCGO ₃	FCGO	FCGO ₃	FCGO
C9H2	50	0.60179831	0.60180164	0.60194158	0.60188559	0.60212767	0.60198610
C9H2	100	0.41000347	0.40951410	0.41262322	0.41032244	0.41430352	0.41069082
C14H2	50	12.2215536	12.2215525	12.2265352	12.2240545	12.2340670	12.2340047
C14H2	100	8.42678910	8.43038094	8.45817834	8.44976407	8.47451242	8.46421674
C18	50	1.20692896	1.20692787	1.20735210	1.20720667	1.20835093	1.20834072
C18	100	0.82023411	0.81999526	0.82244961	0.82160152	0.82402818	0.82253533
C44H1	50	0.78013158	0.78013069	0.78032751	0.78019002	0.78089550	0.78021572
C44H1	100	0.53896055	0.53861740	0.54072973	0.53969664	0.54203137	0.54038731
C59H1	50	0.03268597	0.03268597	0.03268687	0.03268663	0.03268762	0.03268711
C59H1	100	0.02285833	0.02283018	0.02290124	0.02285935	0.02292075	0.02288311
C92H4	50	1.28335254	1.28323674	1.28561541	1.28354938	1.29066447	1.28368185
C92H4	100	0.87626883	0.87167703	0.87863639	0.87378062	0.88152247	0.87551589
C101H2	50	1.03811199	1.03810669	1.03829727	1.03822901	1.03834899	1.03833763
C101H2	100	0.71779457	0.71658293	0.72000932	0.71746358	0.72181785	0.71826797
C171H4	50	0.86673603	0.86580999	0.86764643	0.86632795	0.86855578	0.86694377
C171H4	100	0.59418910	0.59031559	0.59686990	0.59166876	0.59906470	0.59271508
C196H5	50	1.14485651	1.13966297	1.14751234	1.14261785	1.15047581	1.14671974
C196H5	100	0.75617631	0.75171119	0.76214918	0.75417235	0.76747364	0.75618656
C202H3	50	0.69673619	0.69219405	0.69866797	0.69219405	0.69997470	0.69219405
C202H3	100	0.47600555	0.47255797	0.47795849	0.47423304	0.47886663	0.47493585
#Better		2	17	0	20	0	20
#Equal		1	1	0	0	0	0
#Worse		17	2	20	0	20	0

The local optimization procedure of the proposed FCGO algorithm employs a subregion-based acceleration strategy. To further demonstrate its effectiveness, we conducted an experiment using 20 representative instances of the CRICMR problem with center placement constraints. We created an FCGO₃ variant by disabling the acceleration strategy in FCGO while keeping all other components

of the algorithm unchanged. Then, we ran the FCGO and FCGO₃ algorithms independently 10 times for each instance, respectively.

The comparative results of the two algorithms are summarized in Table 1, including the best objective value over 10 runs (R_{best}), the average objective value (R_{avg}) and the worst objective value (R_{worst}). The final three rows indicate the number of instances where the corresponding algorithm outperforms, matches, or underperforms the other. Table 1 shows that the FCGO algorithm significantly outperforms its variant FCGO₃. In terms of R_{best} , FCGO obtains a better, equal, and worse result for 17, 1, and 2 instances, respectively. In terms of both R_{avg} and R_{worst} , FCGO outperforms FCGO₃ for all tested instances. This experiment confirms that the acceleration strategy of the FCGO algorithm plays an important role in its high performance.

1.2. Effectiveness of the perturbation strategy

To jump out of local optima, the FCGO algorithm employs two perturbation operators in a probabilistic way, i.e., the uniform random perturbation and the directed perturbation. To show the effectiveness of this hybrid perturbation strategy, we conducted a comparative experiment with FCGO and its two variants FCGO₄ and FCGO₅, which are obtained by respectively disabling the directed perturbation and the uniform random perturbation while keeping other components of the algorithm unchanged. We ran the three algorithms 10 times for each of the 30 representative instances.

The computational results are summarized in Table 2. Table 2 shows that the hybrid perturbation strategy used in FCGO significantly outperforms the two other strategies, which only use one perturbation operator. Specifically, FCGO, FCGO₄, and FCGO₅ obtain the best results for 16, 12 and 4 instances, respectively, in terms of R_{best} . In terms of R_{avg} , FCGO, FCGO₄, and FCGO₅ obtain the best results for 18, 8 and 4 instances, respectively. This experiment demonstrates that two perturbation operators are functionally complementary and that using them together efficiently enhances the algorithm's search ability.

We provided in Figure 1 the empirical RTD plots of the FCGO algorithms with the three different perturbation strategies for 4 representative instances, where the algorithms were run 100 times for each instance and the target objective values were set to the average objective value of FCGO reported in Table 2. One can observe from Figure 1 that the hybrid perturbation strategy usually has a higher probability of reaching the target objective value under the same computation time compared to the other two perturbation strategies, which further confirms its effectiveness.

Table 2 Comparison between the FCGO algorithms with different perturbation strategies on 30 representative instances, where the best results in bold in terms of R_{best} , R_{avg} and R_{worst} of the compared algorithms are indicated.

Instance		R_{best}			R_{avg}			R_{worst}		
Region	p	FCGO ₄	FCGO ₅	FCGO	FCGO ₄	FCGO ₅	FCGO	FCGO ₄	FCGO ₅	FCGO
C9H2	50	0.60179944	0.60179775	0.60180164	0.60182890	0.60186745	0.60188559	0.60189491	0.60198600	0.60198610
C9H2	100	0.40901329	0.41044747	0.40951410	0.41043409	0.41161153	0.41032244	0.41131144	0.41235067	0.41069082
C9H2	200	0.28198614	0.28216408	0.28158182	0.28294610	0.28316742	0.28209036	0.28369566	0.28377347	0.28264220
C14H2	50	12.2215570	12.2215588	12.2215525	12.2265459	12.2269602	12.2240545	12.2340126	12.2369444	12.2340047
C14H2	100	8.42597948	8.44637017	8.43038094	8.44578046	8.46302963	8.44976407	8.45928056	8.47499078	8.46421674
C14H2	200	5.84531107	5.86376183	5.86117662	5.86600104	5.88656122	5.86990314	5.88205521	5.90416710	5.87700494
C18	50	1.20692894	1.20828377	1.20692787	1.20706737	1.20860334	1.20720667	1.20828638	1.20894040	1.20834072
C18	100	0.81935085	0.82089720	0.81999526	0.82172480	0.82225170	0.82160152	0.82316083	0.82376404	0.82253533
C18	200	0.56633560	0.56747859	0.56697734	0.56736467	0.56856231	0.56821827	0.56938815	0.56957693	0.56985959
C44H1	50	0.78013081	0.78013196	0.78013069	0.78019804	0.78060868	0.78019002	0.78023567	0.78090031	0.78021572
C44H1	100	0.53896332	0.53895648	0.53861740	0.54000296	0.54005111	0.53969664	0.54098178	0.54064634	0.54038731
C44H1	200	0.37620831	0.37646785	0.37588237	0.37713631	0.37684577	0.37641816	0.37760750	0.37734817	0.37672764
C59H1	50	0.03268599	0.03268668	0.03268597	0.03268682	0.03268716	0.03268663	0.03268719	0.03268734	0.03268711
C59H1	100	0.02283292	0.02283329	0.02283018	0.02286385	0.02289196	0.02285935	0.02288915	0.02291981	0.02288311
C59H1	200	0.01537514	0.01540221	0.01536318	0.01539522	0.01544461	0.01537667	0.01541077	0.01546442	0.01538980
C92H4	50	1.28353885	1.28322080	1.28323674	1.28363090	1.28354516	1.28354938	1.28387750	1.28396000	1.28368185
C92H4	100	0.87134018	0.87225020	0.87167703	0.87421412	0.87299189	0.87378062	0.87536129	0.87387661	0.87551589
C92H4	200	0.60388300	0.60392215	0.60362551	0.60499828	0.60512391	0.60483766	0.60606036	0.60582760	0.60674327
C101H2	50	1.03810270	1.03809339	1.03810669	1.03832707	1.03827320	1.03822901	1.03834111	1.03833564	1.03833763
C101H2	100	0.71546709	0.71721698	0.71658293	0.71685826	0.71809859	0.71746358	0.71764947	0.71896871	0.71826797
C101H2	200	0.49465788	0.49579107	0.49549098	0.49670596	0.49644729	0.49668165	0.49815510	0.49740858	0.49798522
C171H4	50	0.86573837	0.86595340	0.86580999	0.86640661	0.86656510	0.86632795	0.86674850	0.86698394	0.86694377
C171H4	100	0.59128774	0.59082005	0.59031559	0.59251707	0.59212405	0.59166876	0.59416705	0.59320498	0.59271508
C171H4	200	0.40662863	0.40775016	0.40597749	0.40893423	0.40837382	0.40823957	0.40914112	0.40918973	0.40904930
C196H5	50	1.13980932	1.13983913	1.13966297	1.14345019	1.14339062	1.14261785	1.14737098	1.14694207	1.14671974
C196H5	100	0.75379835	0.75288980	0.75171119	0.75557371	0.75424507	0.75417235	0.75708645	0.75542476	0.75618656
C196H5	200	0.52062843	0.52166433	0.52027541	0.52259145	0.52268118	0.52252890	0.52397079	0.52350526	0.52385141
C202H3	50	0.69219405	0.69219405	0.69219405	0.69226708	0.69221811	0.69219405	0.69292432	0.69231672	0.69219405
C202H3	100	0.47235596	0.47358522	0.47255797	0.47420241	0.47519442	0.47423304	0.47520564	0.47592500	0.47493585
C202H3	200	0.32473208	0.32534634	0.32523547	0.32552801	0.32678806	0.32574282	0.32636235	0.32745509	0.32649825
#Best		12	4	16	8	4	18	7	6	17

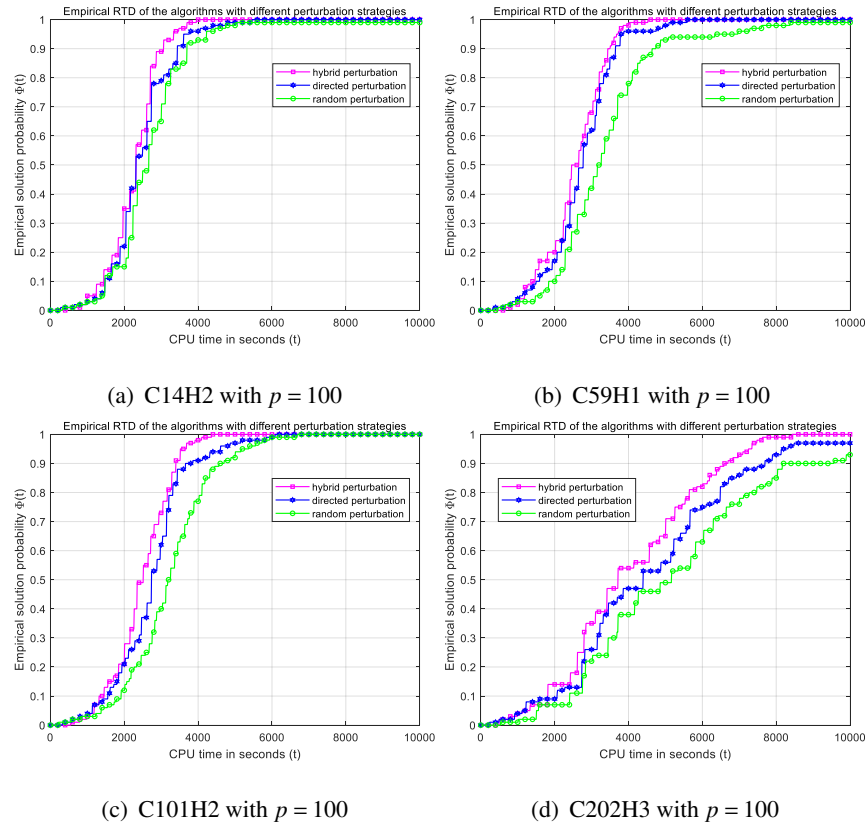


Figure 1 Empirical RTD of the proposed algorithms with different perturbation strategies on 4 representative instances.